Decent Coaching Classes

VIII SB

Mathematics: Chapter 5 - Expansion Formulae

Time: 1 Hour Marks: 30

Q.1 (A) Choose the correct alternative. (5)

1) $(x + 3)(x + 3) = ?$

   a) $x^2 + 6x + 9$ b) $x^2 + 3x + 9$ c) $x^2 + 9$ d) $x^2 - 6x + 9$

2) The expansion of $(a + b)^3$ is:

   a) $a^3 + 3a^2b + 3ab^2 + b^3$ b) $a^3 - 3a^2b + 3ab^2 - b^3$ c) $a^3 + b^3$ d) $a^3 + 3ab + b^3$

3) $(2p + 3m)^2 = ?$

   a) $4p^2 + 12pm + 9m^2$ b) $4p^2 + 9m^2$ c) $4p^2 + 6pm + 9m^2$ d) $2p^2 + 6pm + 3m^2$

4) In the expansion of $(x + a)(x + b)$, the coefficient of x is:

   a) $a + b$ b) $a - b$ c) ab d) $a + b + ab$

5) $(5x - 2y)(5x + 2y) = ?$

   a) $25x^2 - 4y^2$ b) $25x^2 + 4y^2$ c) $25x^2 - 20xy + 4y^2$ d) $25x^2 + 20xy - 4y^2$

Q.2 Attempt any FOUR of the following. (4 × 2 = 8)

1) Expand: $(x + 4)(x - 2)$
2) Expand: $(3x - 2y)(3x + 2y)$
3) Find the value of $(99)^2$ using an expansion formula.
4) Expand: $(2p + q + 3)^2$
5) Expand: $(a + 7)(a - 5)$

Q.3 Attempt any THREE of the following. (3 × 3 = 9)

1) Expand: $(x + 2)^3$
2) Expand: $(m - 3)^3$
3) Find the cube of 102 using an expansion formula.
4) Simplify: $(p + q)^3 + (p - q)^3$

Q.4 Attempt any TWO of the following. (2 × 4 = 8)

1) Expand: $(2x - 3y)^3$
2) Expand: $(a + 2b + 3c)^2$
3) Simplify: $(3x + 4y)^3 - (3x - 4y)^3$

— All the Best —